

Research Article

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A goal based index to analyze the competitive balance of a football league

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Abstract: Competitive balance in a football league is extremely important from the perspective of economic growth of the industry. Many researchers have earlier proposed different measures of competitive balance, which are primarily adapted from standard economic theory. However, these measures fail to capture the finer nuances of the game. In this work, we discuss a new framework which is more suitable for a football league. First, we present a mathematical proof of a theoretically optimal situation where a football league becomes perfectly balanced. Next, a goal based index for competitive balance is developed. We present relevant theoretical results and show how the proposed index can be used to formally test for the presence of imbalance. The methods are implemented on the data from the top five European leagues, and it shows that the new approach can be helpful in explaining the changes in the seasonal competitive balance of the leagues. Further, using panel data models, we show that the proposed index is more suitable to analyze the variability in total revenues of the football leagues. We also discuss how the methods can be easily extended to develop other goal-based indices under different modeling assumptions.

Keywords: concentration ratio; Herfindahl index; panel data; Skellam distribution; soccer.

1 Introduction

Football (or association football, or soccer) is arguably the most popular sport in the world. Especially in Europe, it is not only popular, but also a big industry and is growing every year. In 2018/19 season, the total revenue in Europe's top five leagues (England, France, Germany,

Italy, Spain) was approximately 17 billion Euros, which is more than double what it was a decade ago. In fact, the majority of the total value of the entire European football market is represented by these five leagues. Dima (2015) discussed this market structure in more detail. Rohde and Breuer (2017), meanwhile, elaborated on the history and market situation of these five leagues, and how they have been undergoing significant transformation regarding foreign investment over the last decade or so. Continuing on that note, with respect to the growth of the market, a common hypothesis is that the competitive balance in a sports league, which primarily relies on the strength and talent of the teams, is a crucial factor to affect public interest and thereby the financial health of the industry of that sport. Rottenberg (1956) is a seminal work in this context. This study brought forth the Uncertainty of Outcome Hypothesis (UOH) in the sports economics literature. One can argue that if there is a high degree of uncertainty about the outcomes of a match, it attracts more viewers and thereby more revenue for the teams as well as the league. There have been a few interesting studies over the last couple of decades where the authors have analyzed the effect of competitive balance on attendance, broadcast rights or revenue sharing of the football industry in Europe. One of the most interesting works in this regard was by Buraimo and Simmons (2009) who analyzed the case of Premier League matches, and explored how the market size and level of competition are influencing the fan base. Késenne (2000), Szymanski (2001), Peeters (2011) and Plumley et al. (2018) are some other notable examples in this field of research.

Naturally, it is of primary importance to have a good measure of competitive balance. Szymanski (2003) distinguished among three different ways of measuring competitive balance: match-level, championship-level and season-level. Our focus in this paper is on a season-level measure which can appropriately capture the competitiveness of a league as a whole. One of the earliest and standard techniques in this regard is to use the ratio of standard deviation of actual win percentages in the league to the same in an ideal league where each match can end in equally likely outcomes. While this measure

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and its variants have been used extensively in many studies (see Quirk and Fort (1997) and Humphreys (2002) for example), Michie and Oughton (2004) argued in detail why it is not adequate in the context of football. One of the main limitations of this measure is that it was primarily developed for major sports in the United States of America (such as baseball, basketball etc.) where draws are almost non-existent, but that is not the case for football. For instance, in the 2018-19 season, approximately 23.9, 18.7, 29, 29 and 28.4% matches were drawn respectively in Bundesliga, English Premier League (EPL), La Liga, Ligue 1 and Serie A. In a more recent study, Owen and King (2015) criticized the standard deviation based measure for its sensitivity to the number of teams in the league. Doria and Nalebuff (2021) also scrutinized in detail how the above measure is affected by varying season lengths, and therefore is not suitable to compare different leagues. Another significant drawback of the above measure is that it fails to capture the presence of dominance in the league. In order to circumvent these issues, two other measures have been developed from the standard economic theory. The first one is inspired from the concept of a concentration ratio (see Hall and Tideman (1967)) which quantifies the fraction of industry size held by the top L firms. In case of a football league, one can consider the top L clubs and compute the total proportion of their point share to define the concentration ratio. Note that the first four to six teams (depending on the league) of every league qualify for the European competitions. In view of that, we use $L = 6$ and define the six club concentration index or C6 index as follows.

Definition 1. (C6 index). For a league with n teams, let P_i be the total points gathered by the i th team at the end of the season. The point share of the i th team is defined by $P_i^* = P_i / \sum_{i=1}^n P_i$. Let $P_{(j)}^*$ denote the j th largest value in the set $\{P_1^*, P_2^*, \dots, P_n^*\}$. Then, the C6 index is defined as

$$C6 = \frac{n}{6} \sum_{j=1}^6 P_{(j)}^*. \quad (1.1)$$

The C6 index quantifies the imbalance between the top six clubs in a league and the rest, but it fails to capture the change in imbalance within these two groups. In fact, as Manasis et al. (2011) pointed out, the above formulation of the concentration ratio is affected by the extent of domination as well as different numbers of teams in different leagues. Consequently, it may lead to misleading interpretations. In view of that, we adopt their idea to define the normalized C6 index as a better measure of competitive balance.

Definition 2. (Normalized C6 index). Consider a league with n teams, and assume that every win in the league is awarded with 3 points. Let $P_{(j)}$ be the j th largest value in the set $\{P_1, \dots, P_n\}$, where P_i holds the same meaning as above. Then, the normalized C6 index is defined as

$$NC6 = \frac{1}{18(n-6)} \left[\sum_{j=1}^6 P_{(j)} - 18(n-1) \right]. \quad (1.2)$$

It is vital to note that Manasis et al. (2011) assumed no draws to derive the maximum and minimum bounds for formulating the above normalization (they also assumed 2 points for a win, which we have modified appropriately). For a general scoring system with draws taken into account, further modification can be worked out. We, however, avoid such mathematical elaboration here and use the above expression, since its impact on the main analysis would be minimal.

The second measure we shall discuss in this paper is derived from the Herfindahl-Hirschman index (HHI) and it captures the inequalities between all the clubs in a league.

Definition 3. (Herfindahl index of competitive balance).

Consider a league with n teams, and let P_i^* be the point share of the i th team as defined in Definition 1. Then, the Herfindahl index of competitive balance (HICB) is defined as

$$HICB = n \sum_{i=1}^n P_i^{*2}. \quad (1.3)$$

In essence, HICB is a weighted average index for the league where the weights are proportional to each club's point share. Evidently, it reflects the extent of competitive balance among all the teams. As the competitive balance declines, there is a greater inequality among the teams and hence the value of the above measure increases. The reader is referred to Zimbalist (2002), Brandes and Franck (2007) and the references therein for more relevant discussions. Akin to the earlier discussions, HICB is also susceptible to the number of teams in the leagues and appropriate normalization has been suggested. Owen et al. (2007) showed that the upper bound of the term $(HICB - 1)/n$, under the setting of complete imbalance (assuming no draws) is $(n+1)/3n(n-1)$ and it is achieved when the teams win in cascade, i.e. each team beats all teams below it and is defeated by all teams above it. Avila-Cano et al. (2021), on the other hand, worked under more general assumptions, taking into account the possibility of draws as well as various scoring systems, and derived the upper bound of the Herfindahl index by making use of the truncated cascade distribution. They established that

the most unequal distribution of points, and consequently the maximum value of the index, is achieved when the teams in a league are distributed in two groups of sizes $q(n)$ and $n - q(n)$ such that all teams in the first group win their matches in cascade and against all teams in the second group whereas all teams in the second group draw against each other. In continuation with this result, the authors also derived the corresponding maximum values (see Table 2 in Avila-Cano et al. (2021)). Denoting this maximum value as $m(n)$ and considering the suggestions of the above two papers, we shall use the following normalized version of the index in this article.

Definition 4. (Normalized HICB). For a league with n teams, let HICB be calculated as in Eq. (1.3). Then, the normalized HICB is defined as

$$\text{NHICB} = \left(m(n) - \frac{1}{n} \right)^{-1} \left[\frac{\text{HICB}}{n} - \frac{1}{n} \right]. \quad (1.4)$$

Continuity of performance by the individual teams within a league is another important aspect of competitive balance. Haan et al. (2007) discussed it in depth and have proposed the following measure in the context of football.

Definition 5. (Normalized measure of dynamics). Consider a league of n teams. Let $R_{i,1}$ be the rank of the i th team in a particular season, and let $R_{i,0}$ be the rank of the same team in the previous season. Then, a normalized measure of dynamics of the league for the current season is calculated as

$$\text{ND} = \frac{2}{n^2} \sum_{i=1}^n |R_{i,1} - R_{i,0}|. \quad (1.5)$$

Note that the concentration ratio and the Herfindahl index have been widely used and can be considered to be benchmarks for measuring concentration of firms within an industry. While the adaptation to football leagues is natural and understandable, there is an interesting limitation that most authors have failed to address. In football leagues, it is not rare to have more than one team finishing up with the same number of points, in which case the standing of those teams is usually determined by the overall goal differences, followed by other criteria such as goals scored, head-to-head record etc. The normalized measure of dynamics uses the ranking to evaluate the continuity of performance across seasons, but it fails to furnish an idea about within-season competitiveness. Our objective in this paper is to circumvent these issues and propose a new measure that uses the number of goals scored by the two teams instead of just the outcomes. Our theory is motivated by the idea that perfect balance in a

football league should mean identically distributed points, goal differences, and goals scored for the teams in the entire season. This leads to the following definition of a balanced league.

Definition 6. (Perfectly balanced league). A league is ‘perfectly balanced’ if any permutation of the teams is equally likely to be the final standing.

The paper is structured as follows. Section 2 lays out the theoretical background, along with relevant results and methods to analyze the data. A short simulation study is presented in Section 3. Next, we analyze decade long data from the top five European leagues using the proposed method, and provide a comparative study of various measures in terms of explaining the changes in overall revenues. These results are provided in Section 4. Following that, we summarize the advantages of the proposed approach in Section 5. Some important concluding remarks are also presented in that section.

2 Methods

In the discussions below, n denotes the number of teams in the league. All of them play against each other in a home-and-away system, meaning that the total number of matches is $N = n(n-1)$. Z_{ij1} (correspondingly, Z_{ij0}) is the outcome of the match between the i th and the j th team, played in the home ground of the i th team (correspondingly, the j th team). For instance, $Z_{ij1} = 1, 0, -1$ indicate respectively a win, draw and loss at home for the i th team against the j th team. Further, let P_{ij1} (and P_{ij0}) denote the points gathered by the i th team against the j th team at home (and away); and let X_{ij1} and X_{ij0} be the corresponding number of goals scored. We shall use P_i , GS_i , GC_i and GD_i to denote the overall points, total goals scored, total goals conceded and the overall goal difference of the i th team. Thus,

$$\begin{aligned} P_i &= \sum_{j=1}^n \sum_{k=0}^1 P_{ijk}, \quad GS_i = \sum_{j=1}^n \sum_{k=0}^1 X_{ijk}, \quad GC_i \\ &= \sum_{j=1}^n \sum_{k=0}^1 X_{jik}, \quad GD_i = GS_i - GC_i. \end{aligned} \quad (2.1)$$

Note that, for $i = 1, 2, \dots, n$, the final standing of the league is determined first by P_i , then by GD_i and then by GS_i . This is the standard norm for most domestic leagues and we shall consider this rule throughout the study.

Assumption 1. X_{ijk} , for all $i, j \in \{1, 2, \dots, n\}, k \in \{1, 0\}$, are independent and X_{ijk} follows a Poisson distribution with mean λ_{ijk} .

Our main theory revolves around the above assumption. Note that different values of λ_{ijk} can be attributed to various factors that affect the number of goals scored by a team in a match. See Clarke

and Norman (1995) and Everson and Goldsmith-Pinkham (2008) for related discussions.

Let us start with a theoretically optimal setup in which every outcome of every match is equally likely, thereby making a league perfectly balanced. We emphasize that the following theorem is stated for completeness of the paper, and it provides only a sufficient condition for perfect balance. We shall show later that the assumptions can be further relaxed, which would subsequently lead to our proposed index of competitive balance.

Theorem 1. *If Assumption 1 is true with $\lambda_{ijk} = \lambda > 0$ for all $i, j \in \{1, 2, \dots, n\}, k \in \{1, 0\}$, then there exists a λ_0 such that every match is equally likely to end in a win, draw or loss for any team.*

Detailed proof of the above theorem is relegated to the Appendix for brevity. It leverages the concept of the modified Bessel function of the first kind. An immediate consequence of the above is the following.

Corollary 1. *If X_{ijk} , for all $i, j \in \{1, 2, \dots, n\}, k \in \{1, 0\}$, are iid Poisson(λ_0), then a football league is perfectly balanced.*

The proof is trivial in view of the fact that every match in the league is iid multinomial with the probability of every outcome being $1/3$ and that the total number of goals scored (and conceded) by every team are iid Poisson with mean $2\lambda_0(n-1)$. The latter ensures that in case two or more teams have an equal number of points, they are equally likely to have a better goal difference (similarly, better goals scored, better head-to-head record as well) than others.

Remark 1. The value of λ_0 is computed to be approximately 0.88. Figure 1 further shows how the probability of a draw changes with different values of λ . Note that, under the assumptions of Theorem 1, if the probability of a draw is d_λ , then any team can win or lose with equal probability $(1 - d_\lambda)/2$. From the figure, it can be observed that as $\lambda \rightarrow 0$, a draw is expected whereas for large λ , the chance of having

a definitive result increases. This observation is in the same spirit of Propositions 1 and 2 from Scarf et al. (2019).

Interestingly, λ_0 is not the only value that makes a league perfectly balanced. It ensures that every outcome of a match is equally likely, a sufficient condition for a perfectly balanced league. However, this is not a necessary condition. In fact, the following is true.

Theorem 2. *If every win, draw and loss are awarded 3, 1 and 0 points respectively, and if Assumption 1 is true with $\lambda_{ijk} = \lambda > 0$ for all $i, j \in \{1, 2, \dots, n\}, k \in \{1, 0\}$, then a football league is perfectly balanced.*

The above-mentioned point structure has been the practice in football leagues over the last few decades. A detailed proof is presented in the Appendix. We use properties of a Skellam distribution (Skellam (1946)) and some combinatorial arguments in the proof. We also point out that even if the point structure is different, a similar idea can be used to get similar results. Furthermore, perfect balance can be achieved under other distributional assumptions on X_{ijk} as well.

Corollary 2. *If the probability distribution of (X_{ij1}, X_{ij0}) is such that the distributions of $Y_{ij} = X_{ij1} - X_{ij0}$ (for all i, j) are iid; then a football league is perfectly balanced.*

Proof of the above is straightforward and can be worked out in an identical manner as in Theorem 2. We shall revisit this aspect later in this section to develop other goal-based indices. For now, we keep our focus on the Poisson assumption and move on to developing a new measure of competitive balance.

Note that under the assumption that all X_{ijk} are iid Poisson(λ), the maximum likelihood estimate (MLE) of λ is given by

$$\hat{\lambda} = \frac{1}{2N} \sum_{i=1}^n \sum_{j=1}^n \sum_{k=0}^1 X_{ijk}, \quad (2.2)$$

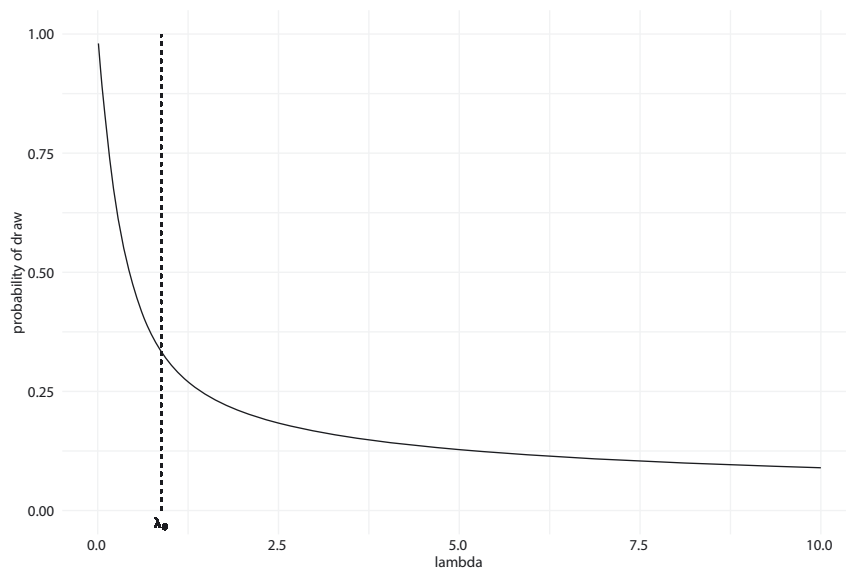


Figure 1: Probability of draw for different values of λ . Ideal situation of $\lambda_0 \approx 0.88$ is displayed with a dotted line.

which is essentially the average number of goals scored in the league. Based on $\hat{\lambda}$, one can also estimate the theoretical probability of draws under Assumption 1. Hereafter, it is denoted by $d_{\hat{\lambda}}$. In Section 4, the values of $\hat{\lambda}$ and $d_{\hat{\lambda}}$ for every league and every season in this study are presented.

Recall that the mean equals the variance for a Poisson distribution. Under the general structure of Assumption 1, the X_{ijk} 's are not identically distributed and are dependent on individual team's skill, opponent, form or the ground. Therefore, the observed variance of the number of goals should be considerably more than the mean. In that light, if we assume that the X_{ijk} 's are independently distributed Poisson random variables, it is the deviations of X_{ijk} 's from $\hat{\lambda}$ which would speak about how much a league deviates from the hypothetical situation of being perfectly balanced. That motivates us to develop the following.

Definition 7. (Goal-based index). Let X_{ij1} (and X_{ij0}) denote the number of goals scored by the i th team against the j th team at home (and away). $\hat{\lambda}$ is the overall mean number of goals in the entire league. Then, the following goal-based index (GBI) is an appropriate measure of competitive balance of the league.

$$GBI = \frac{1}{(2N-1)\hat{\lambda}} \sum_{i=1}^n \sum_{j=1, j \neq i}^n \sum_{k=0}^1 (X_{ijk} - \hat{\lambda})^2. \quad (2.3)$$

Theorem 3. Assume that X_{ijk} , for all $i, j \in \{1, 2, \dots, n\}, k \in \{1, 0\}$, are independently distributed Poisson random variables. Then, for a league perfectly balanced in the sense of Theorem 2, $(2N-1)GBI$ is asymptotically equivalent to a chi-squared random variable with degrees of freedom $2N-1$.

The proof is based on the theory of likelihood ratio tests and is presented in the Appendix. GBI, in comparison to C6 or HICB, is more attractive as it takes into account the number of goals scored by the teams, and not just the individual results of the matches. Moreover, Theorem 3 provides a framework to formally test if a league shows significant deviation from being perfectly balanced or not. One can reject the null hypothesis of perfect balance if $(2N-1)GBI$ is greater than the critical value $\chi^2_{2N-1, \alpha}$, the upper α -quantile of the corresponding chi-squared distribution.

Let us now address the simplicity associated with the independent Poisson assumption which is key to the formulation of the above index. A few studies (such as Karlis and Ntzoufras (2003), Koopman and Lit (2015) and Groll et al. (2018)) argued that the numbers of goals scored by the two teams in a football match are dependent on each other, and therefore a bivariate Poisson model can be more appropriate. On the other hand, Manderson et al. (2018) forecast football match results by using the properties of a Skellam distribution to model goal differences instead of the goals scored. Pelechrinis and Winston (2021) have also used a Skellam regression model to analyze positional value in soccer. Without getting into the debate of which approach is the best among all, we focus on the issue of competitive balance and show below that similar goal based indices can be developed under these assumptions as well. Subsequently, in Section 4, we shall compare the effectiveness of each goal based index in terms of explaining the variation in the revenues.

Definition 8. (Other goal-based indices). Let X_{ijk} (for $i, j = 1, \dots, n$ and $k = 0, 1$) bear the same meaning as in Definition 7.

- (a) Assume that (X_{ij1}, X_{ij0}) follow a bivariate Poisson-lognormal distribution with parameters μ_{ij} and Σ_{ij} , as proposed by Aitchison and Ho (1989). Consider the testing problem with the null hypothesis

$$H_0: \mu_{ij} = \mu, \Sigma_{ij} = \Sigma \quad \text{for all } i, j,$$

against the alternate hypothesis that H_0 is not true.

Let LBP be the likelihood ratio for the above problem where the associated null distribution is $-2 \log LBP \sim \chi_b^2$. Then, the goal-based index based on the above-mentioned bivariate distribution is

$$GBIBP = \frac{-2 \log LBP}{b}. \quad (2.4)$$

- (b) Assume that $(X_{ij1} - X_{ij0})$ follows a Skellam distribution with parameters (λ_i, λ_j) . Consider the testing problem with the null hypothesis

$$H_0: \lambda_i = \lambda_h, \lambda_j = \lambda_a \quad \text{for all } i, j,$$

against the alternate hypothesis H_0 is not true.

Let LS be the likelihood ratio for the above problem such that under H_0 , $-2 \log LS \sim \chi_s^2$. Then, the goal difference based index which relies on the Skellam distribution assumption is

$$GDBI = \frac{-2 \log LS}{s}. \quad (2.5)$$

Remark 2. Calculation of the likelihood ratios in the above cases are standard, and are not presented in detail for the sake of brevity. The degrees of freedom (b, s) in the above two cases are calculated by the differences in the dimensions of $H_0 \cup H_1$ and H_0 , and are therefore dependent on the number of teams in the league. It is also imperative to point out that $Y_{ij} = X_{ij1} - X_{ij0}$, for all i, j , are iid under the null hypotheses in both (a) and (b) of the above definition. Then, following Corollary 2, we can conclude that these setups correspond to the notion of a perfectly balanced league.

We now turn our attention to understand if the proposed goal based indices have better use over NC6, NHICB or ND in explaining the variation of revenue in a league. To that end, we collect the data on season-wide total commercial revenue of the five leagues (source: Deloitte (2020b)). Let $R_{\ell t}$ be the revenue (in billion Euros) of the ℓ th league in the t th season. Let $CB_{\ell t}$ be a measure of competitive balance for the same (we shall apply the following method for all of the above measures). Our primary objective is to assess if $R_{\ell t}$ is significantly dependent on $CB_{\ell t}$. Experimentation suggested that a logarithmic transformation of the revenue renders a better fit for our data. Following Bradbury (2019), we include two other covariates which are known to impact the revenues in similar contexts. First one is the gross domestic product (GDP) of the country corresponding to each league, and it is known to be an apt indicator of a country's economic condition. Second, in order to analyze the effects of stadiums and the market sizes of every league, we use the average season-wide attendance. We shall denote these two variables as $GDP_{\ell t}$ and $Att_{\ell t}$.

It is worth mention that the possibility of including a performance metric at European competitions has also been explored. Although similar covariates have been proved to be effective in other relevant studies (Barajas Alonso et al. 2005; Carmichael, McHale, and Thomas 2011), it did not improve our model and hence is omitted in the interest of parsimony.

Observe that the framework here is suitable for a panel data analysis, and hence, the following model is proposed.

$$\log R_{\ell t} = \alpha_{\ell} + \beta_1 t + \beta_2 \text{GDP}_{\ell t} + \beta_3 \text{Att}_{\ell t} + \gamma \text{CB}_{\ell t} + \epsilon_{\ell t},$$

$$1 \leq \ell \leq 5, 1 \leq t \leq T, \quad (2.6)$$

where $\epsilon_{\ell t}$ are iid errors, α_{ℓ} is the effect of the ℓ th league, β_j coefficients capture the effects of the three covariates and γ is the main parameter of interest as it describes how the measure of competitive balance affects the revenue. For α_{ℓ} , we consider both fixed-effect and random-effect structures and compare their results. In case of random-effect models, α_{ℓ} is taken as $\alpha_0 + \alpha'_{\ell}$ where α_0 is the overall mean, and α'_{ℓ} is the zero-mean random effect for the ℓ th league. After fitting both types of models, we perform a Hausman test (Hausman (1978)) to find out which one works better.

All of the computations in this paper are done in R Studio version 1.2.5033, coupled with R version 3.6.2. The panel data analysis is carried out using the 'plm' package by Millo (2017).

3 Simulation study

In the simulation study, we focus only on GBI and first illustrate the theory from Theorem 1 to Theorem 2 by simulating toy examples. For different values of λ , large number of leagues are generated under Assumption 1. For each league, we calculate the proportion of draws and plot them against the values of λ . Next, we look at the final standing of the league table and calculate the proportion of times each possible permutation is

happening. Under the assumptions we have, Theorem 2 suggests that the league is perfectly balanced and hence every permutation of the teams is equally likely to happen. That indicates a multinomial distribution with equal cell probabilities. On that note, using the observed frequencies of every permutation from the simulated data, we conduct multinomial goodness of fit test and find the corresponding p -values. Note that the number of possible permutations for a league with n teams is $n!$ which is quite large even for small n . So, in order to conduct an appropriate goodness of fit test, one needs to replicate the experiments millions of times. For computational ease and since the objective is to illustrate the results which are already proved, we limit ourselves to $n = 5$ in this case and take 10,000 replications in all experiments.

Figure 2 shows that the proportion of draws decrease steadily as λ increases. The iid assumption of the teams and the games also imply that the proportion of home wins and away wins are expected to be equal, and that is reflected in the left panel of the plot. Right panel of the figure shows the p -values for the multinomial goodness of fit tests based on the 10,000 replications. It is evident that in all cases, the tests fail to reject the null hypothesis of every permutation being equally likely.

Next, we wish to evaluate the effectiveness of GBI in testing whether the goal-scoring patterns of all the teams in the league are iid. We consider leagues of $n = 20$ teams in this case. Once again, for different values of λ , we assume that all X_{ijk} are iid Poisson(λ) and all matches are simulated accordingly. Then, the chi-squared test using GBI, as described in the previous section, is conducted. We use a 5% level of significance throughout this study.

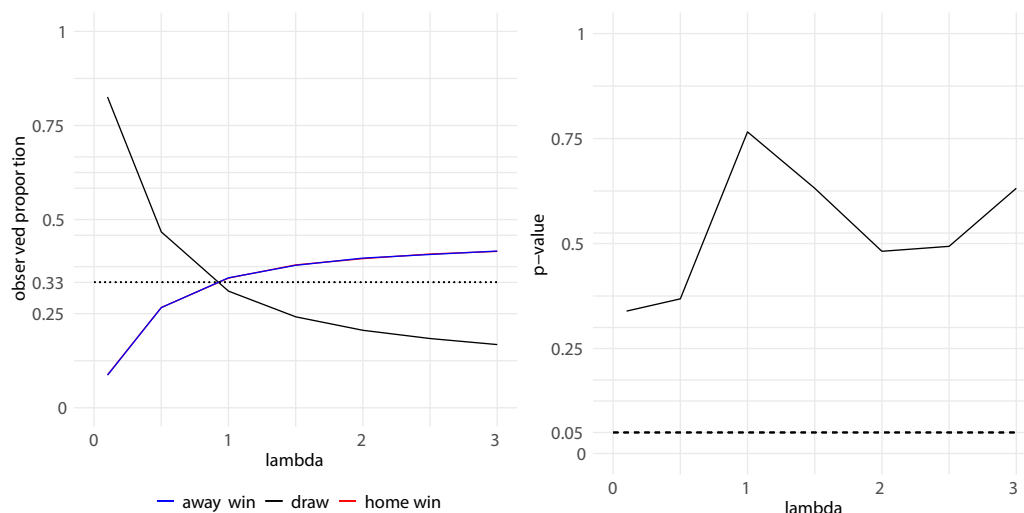


Figure 2: Results from 10,000 simulated leagues under Assumption 1 for different λ : (Left) proportions of the three types of results. Note that the blue and red lines coincide under the assumption. (Right) p -value of the multinomial goodness-of-fit tests.

We replicate the experiment 10,000 times to compute the empirical type I error of the test. These values are displayed in the left panel of Figure 3. We see that for all values of λ , the type I error remains at around 5%. We also compute the power of the test by simulating leagues where all X_{ijk} are not iid. Here, we assume that X_{ijk} are independent and $X_{ijk} \sim \text{Poisson}(1.5 + \lambda_i)$, where λ_i is a team-specific parameter. 20 different scenarios are considered in this regard. In the k th such scenario, λ_i is assumed to be 0 for all but k teams, randomly selected from the 20. For those k teams, the values of λ_i are chosen randomly from a uniform distribution between -1.5 and 1.5 . Note that the value 1.5 is typically the average number of goals scored in various leagues and therefore, this setting corresponds to a realistic scenario for a football league. Next, the GBI is used to detect deviation from the ideal situation. Similar to before, each experiment is repeated 10,000 times and we use them to calculate the power, which is presented in the right panel of Figure 3. It is evident that the testing procedure performs well, recording more than 80% power, when at least seven teams have a different goal-scoring pattern.

4 Application

As mentioned above, we consider ten seasons' data from the top five leagues of Europe – Bundesliga, EPL, La Liga, Ligue 1 and Serie A. These data are collected from the public repository Datahub (link: <https://datahub.io/>). Data on the GDP are obtained from another public repository The

World Bank (link: <https://data.worldbank.org/>), while the attendance information are taken from Deloitte (2020a).

In view of Theorem 1 and the discussions following that, we begin our analysis by computing $\hat{\lambda}$ (see Eq. (2.2)) for every season and every league. Corresponding theoretical probabilities of draws under the assumptions of this study and the observed proportions of drawn matches are also computed. These quantities are denoted as $d_{\hat{\lambda}}$ and $\hat{D}$, respectively. Overall, the hypothesized probabilities are close to the observed proportions of draws. There are a few cases, for example 2013-14 and 2018-19 seasons of EPL, 2010-11 season of La Liga and Ligue 1, and 2014-15 season of Serie A, when the difference between $d_{\hat{\lambda}}$ and $\hat{D}$ is more than or equal to 0.05. These results are shown in Table 1.

Next, the three types of goal-based indices are computed for every league, across the ten seasons, and are compared against the other three benchmark measures. These quantities are displayed in Figure 4. For the GBI values, we also test whether the league significantly deviates from the hypothetical assumption of perfect balance. In the graph, a black circle indicates that the league was not perfectly balanced in that season whereas a black triangle implies that there was not enough evidence to suggest deviation from perfect balance. Similar tests have been run for GBIBP and GDBI as well. We omit those results from the figure, since our focus is primarily on GBI whereas the other two measures are extensions to show the generalizability of the proposed methodology.

There are a couple of interesting aspects that arise out of this plot. Bundesliga, EPL and La Liga, barring a single season for each, have never been perfectly balanced. For

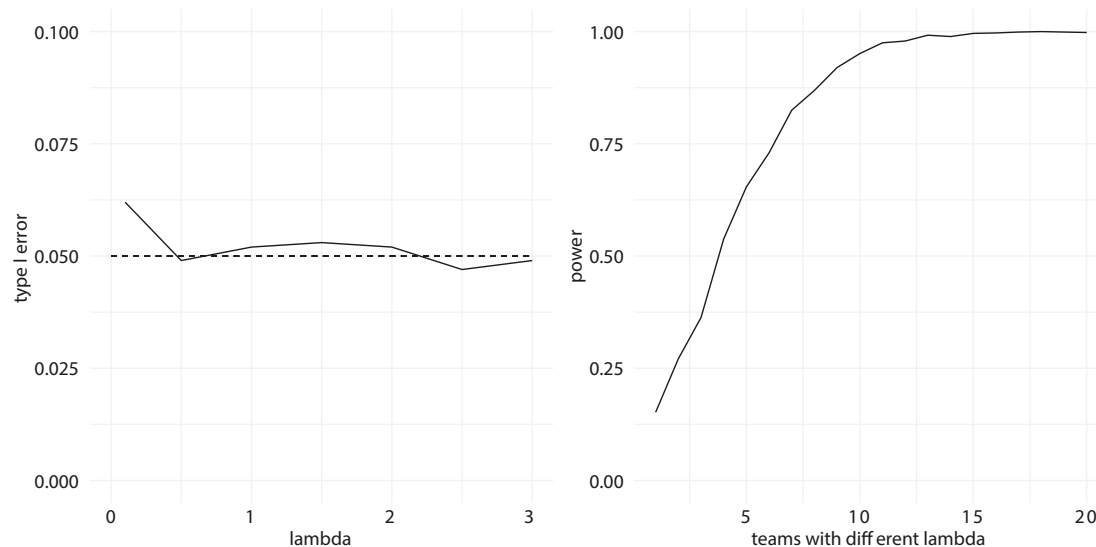


Figure 3: Results on the GBI based test: (Left) type I error from 10,000 simulated leagues under Assumption 1 for different λ ; (Right) power from 10,000 simulated leagues with some number of teams having different scoring pattern.

Table 1: Comparison of theoretical probabilities and the true proportions of draw for every league and every season. $\hat{\lambda}$ is the average number of goals, $d_{\hat{\lambda}}$ is the corresponding theoretical probability, and $\hat{D}$ is the true proportion of draws.

League		2009–10	2010–11	2011–12	2012–13	2013–14	2014–15	2015–16	2016–17	2017–18	2018–19
Bundesliga	$\hat{\lambda}$	1.42	1.46	1.43	1.47	1.58	1.38	1.42	1.43	1.40	1.59
	$d_{\hat{\lambda}}$	0.25	0.25	0.25	0.25	0.24	0.26	0.25	0.25	0.25	0.24
	$\hat{D}$	0.28	0.21	0.26	0.26	0.21	0.27	0.23	0.24	0.27	0.24
EPL	$\hat{\lambda}$	1.39	1.40	1.40	1.40	1.38	1.28	1.35	1.40	1.34	1.41
	$d_{\hat{\lambda}}$	0.25	0.25	0.25	0.25	0.25	0.27	0.26	0.25	0.26	0.25
	$\hat{D}$	0.25	0.29	0.24	0.28	0.20	0.24	0.28	0.22	0.26	0.19
La Liga	$\hat{\lambda}$	1.36	1.37	1.38	1.44	1.38	1.33	1.37	1.47	1.35	1.29
	$d_{\hat{\lambda}}$	0.26	0.26	0.25	0.25	0.26	0.26	0.26	0.25	0.26	0.26
	$\hat{D}$	0.25	0.21	0.25	0.22	0.23	0.24	0.24	0.23	0.23	0.29
Ligue 1	$\hat{\lambda}$	1.21	1.17	1.26	1.27	1.23	1.25	1.26	1.31	1.36	1.28
	$d_{\hat{\lambda}}$	0.28	0.28	0.27	0.27	0.27	0.27	0.27	0.26	0.26	0.27
	$\hat{D}$	0.26	0.34	0.28	0.28	0.28	0.23	0.28	0.25	0.25	0.29
Serie A	$\hat{\lambda}$	1.31	1.26	1.28	1.32	1.36	1.35	1.29	1.48	1.34	1.34
	$d_{\hat{\lambda}}$	0.26	0.27	0.27	0.26	0.26	0.26	0.27	0.25	0.26	0.26
	$\hat{D}$	0.27	0.26	0.29	0.25	0.24	0.32	0.25	0.21	0.22	0.28

Bundesliga, the only season that did not deviate significantly from the assumption of null hypothesis was 2017/18. We find that it was a season when three teams (Schalke, Hoffenheim and Borussia Dortmund) finished on 55 points each, and the qualification for UEFA Champions League had to be determined on goal difference. Meanwhile, in the 2010-11 season of EPL, Chelsea and Manchester City finished second and third with the same number of points, and there were as many as 12 teams within a range of 10 points (39–49). Our approach picks these phenomena nicely and determines that this season was balanced. On the other hand, La Liga has usually observed a higher GBI value, thereby pointing to less competitiveness within the league. The only exception was the 2018-19 season, when three pairs of teams (Getafe and Sevilla, Espanyol and Athletic Bilbao, Valladolid and Celta Vigo) finished on equal points while three more teams (Real Sociedad, Real Betis, Alaves) finished on exactly the same points as well.

So far as Ligue 1 is concerned, it has not observed competitive balance after the 2013-14 season. It might be interesting that this happened shortly after the huge investment on Paris Saint-Germain (PSG) by Qatar Sports Investments, the new majority shareholders of the club. Since 2013-14, PSG has topped the league in all but the 2016-17 season whereas in the six Ligue 1 seasons before 2013-14, there were six different winners. Not only PSG, AS Monaco FC also spent a huge amount after Russian billionaire Dmitry Rybolovlev bought two-thirds of the share and the club secured promotion back to the top

division in 2013. The disparity in spending by these few clubs might have caused the earlier competitive balance to decrease for Ligue 1. Serie A, on the contrary, has been the most balanced league of the lot. In only four out of the ten seasons, their GBI index is significantly higher and there is no discernible pattern in the season-by-season values.

We further notice that the NC6 and NHICB generally behave in the same directions. The overall correlation coefficient between them is 0.92. The correlation between GBI and NC6 as well as between GBI and NHICB is 0.52. Surprisingly, ND is negatively correlated with the two common measures of competitive balance, and shows very different patterns than the other two goal based indices as well. Looking at the earlier graph, we can infer that, except for Bundesliga, the results for ND are not typically consistent with the other measures. GBI shows high positive correlations with both GBIBP and GDBI and one can hypothesize that these newly proposed indices should be equally useful from a practical standpoint. All of the pairwise correlation coefficients between different measures of competitive balance are presented in Figure 5.

The final piece of the analysis is to analyze the relationship between revenue and different types of measures. Figure 6 shows the revenues for every season for the five leagues. The growths of the overall revenues are similar for all the leagues, and that justifies the use of the linear trend term in our panel data model. It is clear that EPL has been

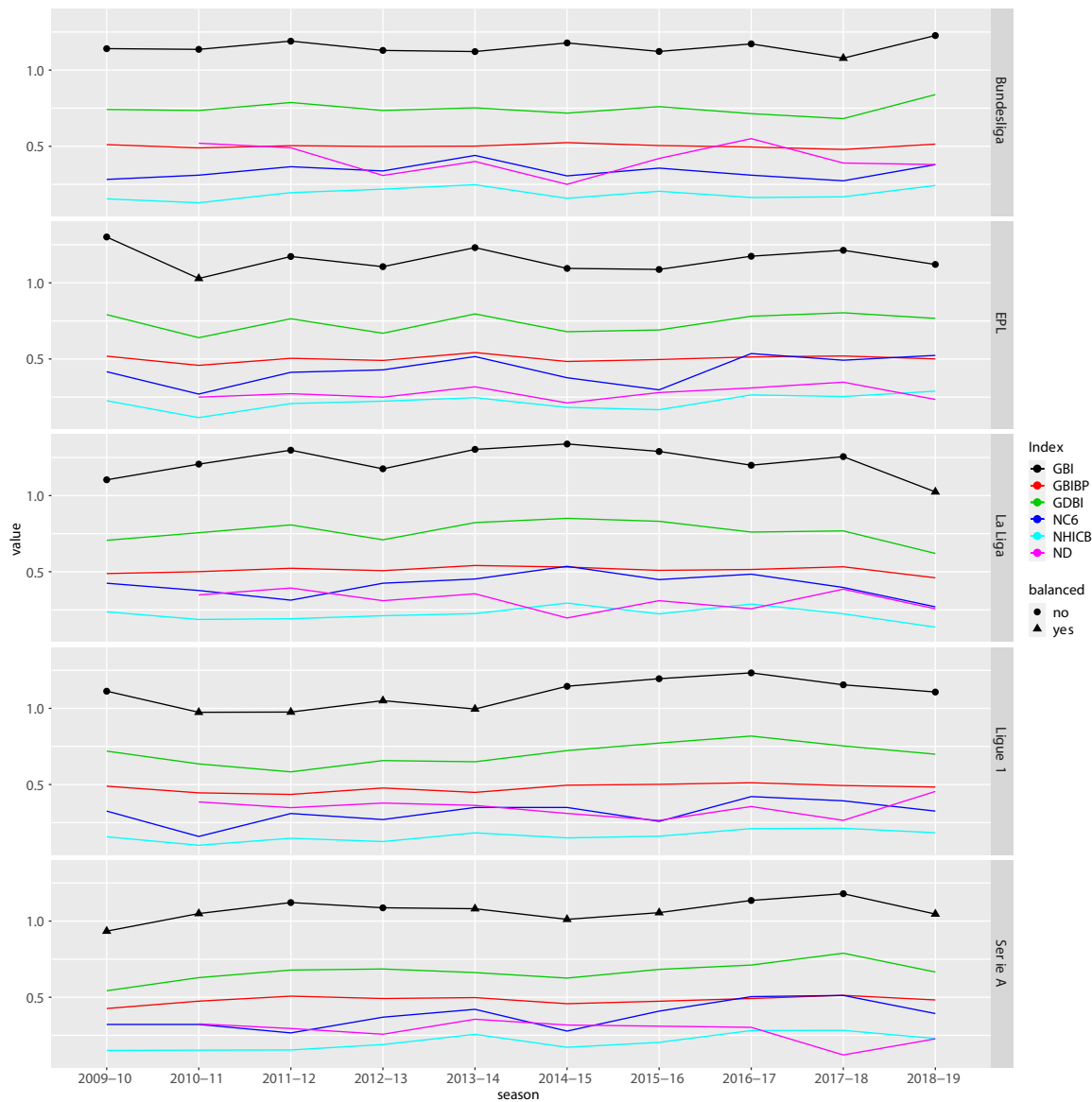


Figure 4: Different measures of competitive balance for the five leagues in ten seasons. GBI values are marked with a circle or a triangle, depending on whether the values significantly deviate from the assumption of perfect balance or not.

the biggest league in terms of revenues generated over the years. Ligue 1 sits at the bottom of the list.

As mentioned before, panel data models (see Eq. (2.6)) are used for the statistical analysis. For each of the three measures, a fixed-effect model (denoted as FE) and a random-effect model (denoted as RE) are implemented on the data. For the RE models, we use the method of Swamy and Arora (1972). In the interest of space, we first present (in Table 2) the results of the Hausman test which is used to choose between FE and RE models in each case.

We observe that the p -values from the Hausman tests between the FE and RE models are non-significant for all cases. It implies that the RE models are providing better

fits for the data for all measures. Next, the results for the six chosen models corresponding to the six measures are displayed in Table 3. There, σ indicates the estimate of the idiosyncratic standard deviation of the model, that is the standard deviation of the error process for a random-effect model; and σ_α is the estimated standard deviation for the random effect.

Looking at the adjusted R^2 values, it is quite evident that the model using GBI performs the best among all measures. We observe that only in the models with GBI and with ND (first and last columns in the table) are the coefficients corresponding to the competitive balance variable significant. However, the RE model with ND offers

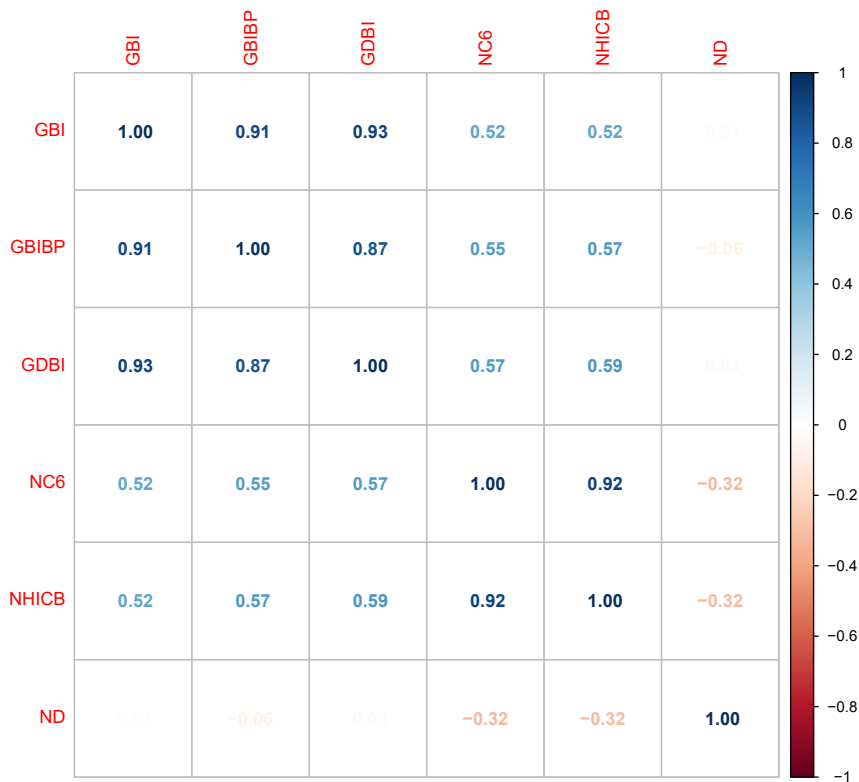


Figure 5: Correlation coefficients between different measures of competitive balance for the entire data.

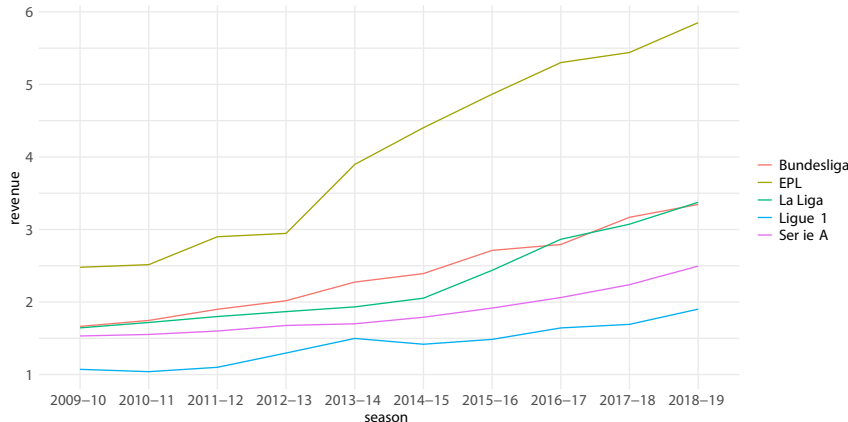


Figure 6: Season-wise revenue (in billion Euros) of the top five European leagues.

the worst fit among all. The linear trend term in each model is significant and the estimates are quite similar across different models. Both attendance and GDP are found to have a positive impact on the revenue, but usually the coefficients are non-significant at the 5% level of significance. Further, note that the standard deviation for the random effect is much higher than the idiosyncratic standard deviation for all RE models. All in all, the RE model with GBI has proved to be the best model here.

A brief discussion of the effect of GBI in the model is of essence here. Recall that a higher value of the measure is equivalent to less competitive balance. Thus,

a significantly negative coefficient for GBI implies that the revenue decreases as a league becomes less competitive. Taking the logarithmic transformation of the response variable into account, the estimated coefficient indicates that with all other factors unchanged, if GBI decreases by an amount of 0.1 (i.e. approximately one standard deviation improvement in the competitive balance), then the revenue would multiply by $\exp(0.0323) \approx 1.033$. In other words, the revenue is estimated to increase by more than 3%. To further elaborate on this note, in Figure 7, we present the estimated season-wise revenues for the five leagues, for different values of GBI. Darker shades in the

Table 2: Results of Hausman tests for fixed effects (FE) versus random effects (RE) models corresponding to different measures of competitive balance.

	Test statistic	<i>p</i> -value	Preferred model
GBI	0.71	0.95	RE
GBIBP	0.64	0.96	RE
GDBI	0.94	0.92	RE
NC6	4.28	0.37	RE
NHICB	2.07	0.72	RE
ND	0.08	0.99	RE

Table 3: Results from the panel data models (RE denotes a random effect model) for different measures of competitive balance, where the dependent variable is the logarithm of revenue. Standard errors (clustered) of the coefficient estimates are given in parentheses. σ and σ_α are the idiosyncratic standard deviation and the random effect standard deviation for the RE models.

	RE(GBI)	RE(GBIBP)	RE(GDBI)	RE(NC6)	RE(NHICB)	RE(ND)
(Intercept)	0.133 (0.468)	0.090 (0.588)	−0.034 (0.438)	−0.374 (0.340)	−0.299 (0.349)	−0.420 (0.357)
Year	0.076*** (0.008)	0.076*** (0.008)	0.077*** (0.008)	0.076*** (0.008)	0.077*** (0.007)	0.074*** (0.012)
Attendance	0.160 (0.096)	0.162 (0.100)	0.171 (0.093)	0.182* (0.085)	0.177* (0.086)	0.167 (0.112)
GDP	0.048 (0.068)	0.051 (0.085)	0.054 (0.077)	0.106 (0.080)	0.082 (0.087)	0.244* (0.119)
GBI	−0.323* (0.125)					
GBIBP		−0.678 (0.562)				
GDBI			−0.345 (0.203)			
NC6				−0.185 (0.141)		
NHICB					−0.364 (0.210)	
ND						−0.975* (0.471)
Adjusted R^2	0.909	0.898	0.898	0.889	0.896	0.794
σ	0.074	0.077	0.075	0.077	0.077	0.072
σ_α	0.386	0.390	0.340	0.171	0.230	0.000
Shapiro <i>p</i> -value	0.629	0.291	0.452	0.072	0.176	0.561

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

bottom-right corners of all plots reflect the significance of the trend pattern as well as the proposed measure. In particular, we draw the reader's attention to the top panel. The estimated values show that the revenue generation in Bundesliga could have been similar to EPL in the last few seasons, if it was more competitive and balanced. While our model has the simplicity of assigning common trend functions to all leagues, these observations still stipulate that our proposed measure can be useful in such analysis.

For the completeness of the above analysis, we next carry out residual diagnostics for the above six models. We primarily assess if the iid normality assumption for the residuals from the models can be considered to be true or not. Shapiro–Wilk normality tests and Anderson–Darling tests are conducted in this regard. The results are analogous and we show the *p*-values for the former in the last row of Table 3. It demonstrates that the models are adequate for all cases. Last but not least, Ljung–Box tests were also conducted to check for autocorrelations among

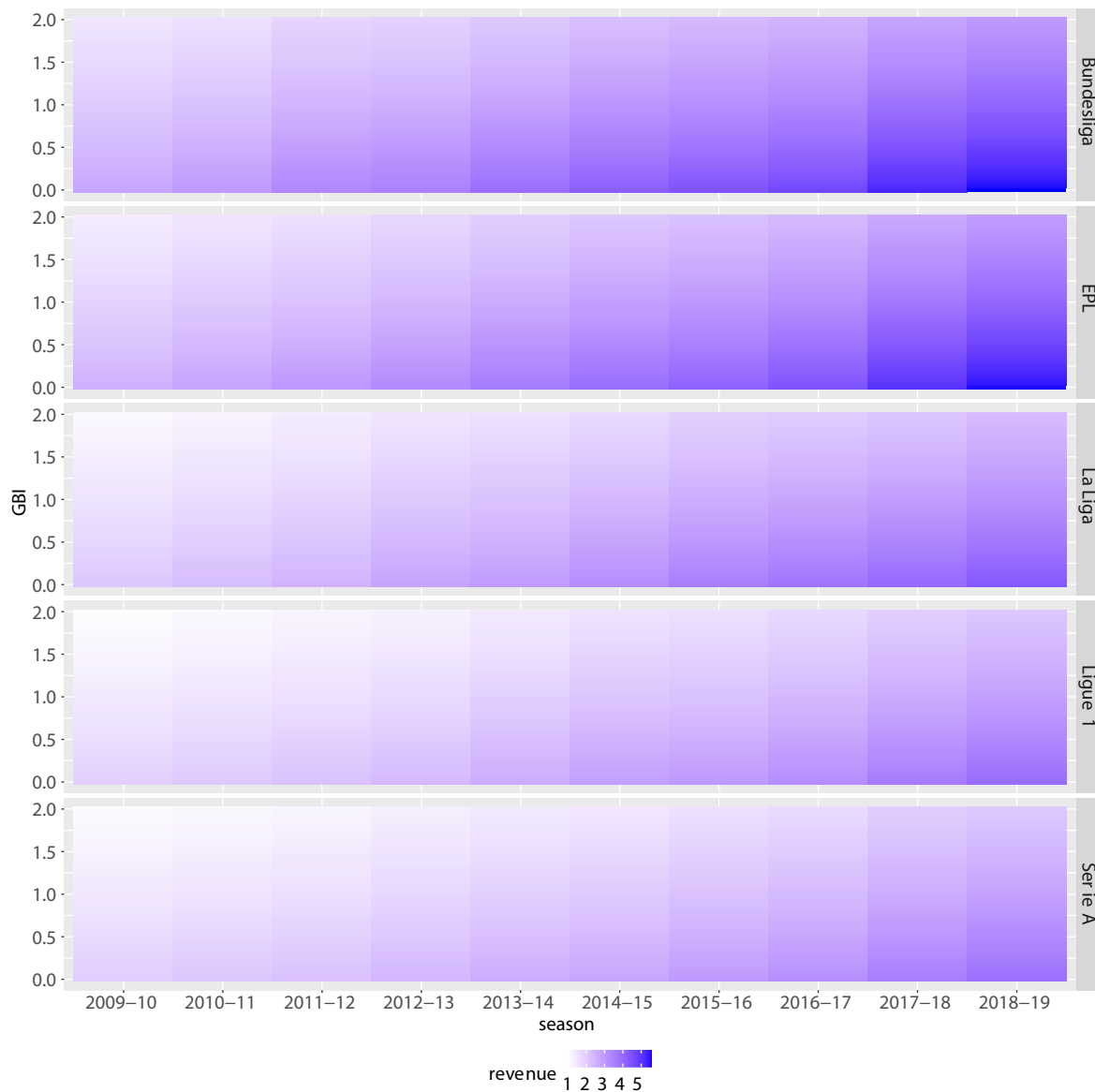


Figure 7: Estimated season-wise revenue (in billion Euros) of the top five European leagues, for different values of GBI.

the residuals within a league and the results confirmed that no significant autocorrelation was present for the proposed model.

5 Concluding remarks

In this study, we have developed a new measure of competitive balance for a football league and have showed that it is more suitable than the existing approaches on various accounts. Earlier measures of competitive balance, such as the concentration ratio or the Herfindahl index, were primarily developed to analyze the balance between firms in an industry; but a football league is more interesting

because of some specific sets of rules to determine the ordering of the clubs. In this work, we have discussed theoretically optimal assumptions under which a league becomes perfectly balanced when we take into account the actual rules of the leagues. To the best of our knowledge, it is the first paper to provide such a mathematical understanding. As the index we propose relies on the number of goals scored by the two teams instead of only the outcomes of the matches, it has a natural advantage over other benchmark techniques. Moreover, a solid theoretical framework behind the GBI makes it possible for us to formally test if a league is perfectly balanced or not. Simulation study confirms that the proposed test has good power.

Extending the proposed approach, we show that similar goal-based indices can be formulated under other modeling assumptions such as bivariate Poisson models and Skellam regression models. In fact, as we analyze the revenue data from different leagues and different seasons, all of these goal-based indices are found to provide better fits than benchmark measures. Adopting similar strategies, one can extend the methodology further to other modeling frameworks as well.

Using the data from the top five European leagues, we also show that GBI can provide some valuable insights about the fluctuations in the competitive balance for a particular league. An interesting observation was the case of Ligue 1, where we see that the league starts to be imbalanced after PSG and Monaco got huge funding. Potentially, one can leverage the GBI to further analyze how investments or player transfers affect the competitive balance of the league. One can also hypothesize that the imbalance in a league results in significant changes in revenue sharing or broadcasting rights. A formal analysis with a bigger dataset in this regard is suitable for a follow-up study to the current work.

On the other hand, we have quantified the seasonal imbalance and provided relevant results. A natural extension is to develop match-level and gameweek-level measures of competitive balance; and to explore how that can be used to predict the final standing of the season. It can be possible to analyze the short-term impact of the gameweek-level competitive balance, for instance on the attendance or television viewership of every match, as well. The methodology can also be easily extended to assess the effects of competitive balance for larger-revenue markets and smaller-revenue markets corresponding to a football league. One can further look at the variation of competitive balance from the championship or the relegation perspective, especially towards the end of the season, and explore its impact on the popularity, viewership, revenue etc.

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Appendix

Proof of Theorem 1.

Let us consider a game where the i th team is playing at home against the j th team. For convenience, let us use Y_1 and Y_2 to denote X_{ij1} and X_{ji0} , respectively. First, we consider the probability of draw, i.e. $\mathbb{P}(Y_1 = Y_2 | \lambda)$. Using the independence assumption and the Poisson probability mass function,

$$\mathbb{P}(Y_1 = Y_2 | \lambda) = \sum_{k=0}^{\infty} \frac{e^{-2\lambda} \lambda^{2k}}{(k!)^2}. \quad (\text{A.1})$$

Note that the modified Bessel function of the first kind of order α is defined by

$$I_{\alpha}(x) = \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(k + \alpha + 1)} \left(\frac{x}{2}\right)^{2k + \alpha}. \quad (\text{A.2})$$

Clearly, Eq. (A.1) can be rewritten as $\mathbb{P}(Y_1 = Y_2 | \lambda) = e^{-2\lambda} I_0(2\lambda)$. As $\lambda \rightarrow 0$, this probability goes to 1. For large z , $I_0(z)$ is approximately equal to $z^{-1/2} e^z / \sqrt{2\pi}$ and therefore, $\mathbb{P}(Y_1 = Y_2 | \lambda) \rightarrow 0$ as $\lambda \rightarrow \infty$. Using the continuity of both terms, it is easy to argue that there exists a $\lambda_0 > 0$ such that the probability of draw is $1/3$. Finally, using the iid assumption, $\mathbb{P}(Y_1 > Y_2 | \lambda_0) = \mathbb{P}(Y_1 < Y_2 | \lambda_0)$ and that completes the proof. $\square$

Proof of Theorem 2. Let $Y_{ij} = X_{ij1} - X_{ji0}$. Clearly, $Y_{ij} > 0$ indicates a home win for the i th team, $Y_{ij} < 0$ indicates an away win for the j th team and $Y_{ij} = 0$ corresponds to a draw. Since all X_{ijk} 's are iid Poisson(λ), Y_{ij} 's are iid and follow Skellam distributions (Skellam (1946)) with parameters (λ, λ) . Using the modified Bessel function of the first kind (see Eq. (A.2)), the probability mass function (pmf) of this distribution can be written as

$$\mathbb{P}(Y_{ij} = k) = e^{-2\lambda} I_{|k|}(2\lambda), \quad (\text{A.3})$$

and the corresponding moment generating function (MGF) is

$$M(t) = \exp(\lambda(e^t - e^{-t})^2). \quad (\text{A.4})$$

It is evident that the distribution of Y_{ij} is symmetric. Let $\mathbb{P}(Y_{ij} = 0) = e^{-2\lambda} I_0(2\lambda)$ be denoted as d_{λ} . Then, the probability of winning for any team in any match is $(1 - d_{\lambda})/2 = w_{\lambda}$, say. Thus, the pmf of P_{ijk} is

$$\mathbb{P}(P_{ijk} = 0) = w_{\lambda}, \quad \mathbb{P}(P_{ijk} = 1) = d_{\lambda}, \quad \mathbb{P}(P_{ijk} = 3) = w_{\lambda}, \quad (\text{A.5})$$

and $\mathbb{P}(P_{ijk} = r) = 0$ otherwise.

Without loss of generality, consider any two teams and let S_1, S_2 be their total points in the entire league. One can write $S_1 = S'_1 + S_{12}$, where $S_{12} = S_{121} + S_{120}$, and S'_1 is the total points gathered against all but the 2nd team. Similarly, write $S_2 = S'_2 + S_{21}$. Under the given assumptions, it is easy to note that

$$\mathbb{P}(S_{12} = a, S_{21} = b) = \begin{cases} w_\lambda^2 & \text{for } a = 6, b = 0, \text{ or } a = 0, b = 6, \\ 2w_\lambda d\lambda & \text{for } a = 4, b = 1, \text{ or } a = 1, b = 4, \\ 2w_\lambda^2 & \text{for } a = 3, b = 3, \\ d_\lambda^2 & \text{for } a = 2, b = 2, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{A.6})$$

Denote the support of (S_{12}, S_{21}) by $B \subset A \times A$, with $A = \{0, 1, 2, 3, 4, 6\}$. Then,

$$\begin{aligned} \mathbb{P}(S_1 > S_2) &= \mathbb{P}(S'_1 - S'_2 > S_{21} - S_{12}) \\ &= \sum_{(a,b) \in B} \mathbb{P}(S'_1 - S'_2 > b - a) \mathbb{P}(S_{12} = a, S_{21} = b). \end{aligned} \quad (\text{A.7})$$

In view of the iid nature of every game, S'_1 and S'_2 are iid random variables. Therefore, $\mathbb{P}(S'_1 - S'_2 > -r) = \mathbb{P}(S'_2 - S'_1 > -r) = 1 - \mathbb{P}(S'_1 - S'_2 \leq r)$, and thereby $\mathbb{P}(S'_1 - S'_2 > -r) + \mathbb{P}(S'_1 - S'_2 > r) = 1 - \mathbb{P}(S'_1 - S'_2 = r)$. It further implies the following.

$$\begin{aligned} \mathbb{P}(S_1 > S_2) &= w_\lambda^2(1 - \mathbb{P}(S'_1 - S'_2 = 6)) \\ &\quad + 2w_\lambda d\lambda(1 - \mathbb{P}(S'_1 - S'_2 = 3)) \\ &\quad + (2w_\lambda^2 + d_\lambda^2)\mathbb{P}(S'_1 = S'_2). \end{aligned} \quad (\text{A.8})$$

A straightforward implication of the above, considering iid nature of S'_1 and S'_2 , is that $\mathbb{P}(S_1 > S_2) = \mathbb{P}(S_1 < S_2)$. From Eq. (A.6), one should also note that $\mathbb{P}(S_{12} > S_{21}) = \mathbb{P}(S_{21} > S_{12})$.

Next, recall the definitions of GS_1, GS_2, GD_1, GD_2 from Eq. (2.1). Let $GS_1 = GS'_1 + GS_{12}$, $GS_2 = GS'_2 + GS_{21}$, $GD_1 = GD'_1 + GD_{12}$ and $GD_2 = GD'_2 + GD_{21}$, defined on an identical fashion as above. It is easy to see that GS'_1, GS'_2 are iid Poisson($m\lambda$) for $m = 2n - 4$, GS_{12} and GS_{21} are iid Poisson(2λ), $GD_{12} = GS_{12} - GS_{21}$ follows a Skellam distribution with parameters $(2\lambda, 2\lambda)$, and $GD_{21} = -GD_{12}$. Thus, $\mathbb{P}(GD_1 > GD_2) = \mathbb{P}(GD'_1 - GD'_2 > GD_{21} - GD_{12}) = \mathbb{P}(V + 2GD_{12} > 0)$, where

V follows a Skellam distribution with parameters $(m\lambda, m\lambda)$. Clearly, both V and GD_{12} are symmetric around 0 and hence, simple calculations can prove that $\mathbb{P}(GD_1 > GD_2) = \mathbb{P}(GD_2 > GD_1)$. In a similar line, one can also show that $\mathbb{P}(GS_1 > GS_2) = \mathbb{P}(GS_2 > GS_1)$.

Combining the above results, we can easily argue that the final standing of the two teams is equally likely to be (1, 2) or (2, 1), if they are determined sequentially by (S_1, S_2) , (GD_1, GD_2) , (GS_1, GS_2) and (S_{12}, S_{21}) . In fact, the same can be proved for any other criteria which relies on the iid assumption.

As a last piece of the proof, note that if there are two or more permutations of the teams which are not equally probable to be the final standing, then there are two teams which are not equally likely to be above each other in the league, a contradiction to what we have proved above. $\square$

Proof of Theorem 3. X_{ijk} , for all $i, j \in \{1, 2, \dots, n\}, k \in \{1, 0\}$, are independent. Let X_{ijk} be Poisson with mean λ_{ijk} . Now, consider the null hypothesis $H_0: \lambda_{ijk} = \lambda$ for all $i, j \in \{1, 2, \dots, n\}, k \in \{1, 0\}$, against the alternate hypothesis $H_1: \lambda_{ijk}$'s are not all equal. The joint likelihood of X_{ijk} 's can be written as

$$L(\lambda_{ijk}; i, j \in \{1, 2, \dots, n\}, k \in \{1, 0\}) = \prod_{i=1}^n \prod_{j=1}^n \prod_{k=0}^1 \prod_{i \neq j} \frac{e^{-\lambda_{ijk}} \lambda_{ijk}^{X_{ijk}}}{X_{ijk}!}. \quad (\text{A.9})$$

Let Λ be the likelihood ratio. It is easy to see that the MLE of λ_{ijk} under H_0 is $\hat{\lambda}$ from Eq. (2.2), and the MLE of λ_{ijk} under $H_0 \cup H_1$ is X_{ijk} . Then, the standard likelihood ratio test statistic is

$$\begin{aligned} -2 \log \Lambda &= -2 \sum_{i=1}^n \sum_{j=1}^n \sum_{k=0}^1 \left[(X_{ijk} - \hat{\lambda}) + X_{ijk} \log \left(\frac{\hat{\lambda}}{X_{ijk}} \right) \right] \\ &= 2 \sum_{i=1}^n \sum_{j=1}^n \sum_{k=0}^1 X_{ijk} \log \left(\frac{X_{ijk}}{\hat{\lambda}} \right). \end{aligned} \quad (\text{A.10})$$

Take $f(x) = x \log(x/x')$ for a fixed x' . The Taylor series expansion of $f(x)$ about x' is

$$f(x) = (x - x') + \frac{1}{2x'}(x - x')^2 + \delta,$$

where δ is negligible under H_0 and for a large sample, which is ensured considering $N = O(n^2)$. Thus, under that scenario, the likelihood ratio test statistic can be

approximated as

$$-2 \log \Lambda = 2 \sum_{i=1}^n \sum_{j=1, j \neq i}^n \sum_{k=0}^1 \left[\left(X_{ijk} - \hat{\lambda} \right) + \frac{\left(X_{ijk} - \hat{\lambda} \right)^2}{2\hat{\lambda}} \right]. \quad (\text{A.11})$$

Since the first term above adds up to 0, it is easy to see that $-2 \log \Lambda$ is asymptotically equivalent with $(2N - 1)\text{GBI}$. Finally, the required result follows from the well known theory about likelihood ratio tests that $-2 \log \Lambda \sim \chi^2$ with degrees of freedom $2N - 1$. $\square$

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